

IOT Based Fire Detection and Alert System

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ABSTRACT

Fire accidents are disastrous to life and property because they are detected too late and slow reactions by the emergency services. In order to address this issue, this paper gives a Fire Detection and Alert System based on the Internet of Things which offers round the clock real-time supervision in homes, businesses and factories. The system is a combination of MQ-2 smoke and gas sensor, dedicated gas sensor, DHT22 temperature sensor, and the ESP32 microcontroller to ensure that there is appropriate environmental monitoring. Upon sensing abnormal smoke emission, hazardous gas emission, or sudden increase in temperature, the system remotely initiates a multi-layered response, such as the activation of a buzzer and red LED indicators to bring attention to the people on the ground, activation of a built-in water sprinkler to extinguish the fire at an early stage, and notifying the community around the area, using mobile communication. There is also an additional manual push button that can be used to quiet down false alarms so that it may operate with control. This system was designed in the Arduino IDE and it is designed in a cost-effective and modular manner to make it reliable, easy to maintain, and easy to deploy. To a great extent, detection, suppression, and emergency communication are combined in order to enhance safety. The solution, which has been proposed, will strengthen fire protection measures by allowing the identification of hazards early and taking prompt action. It offers an effective and applicable process of improving the level of safety in various environments.

Keywords— Internet of Things, Fire Detection System, ESP32 Microcontroller, MQ-2 Sensor, DHT22 Sensor, Gas Sensor, Water Sprinkler System, GSM Alert System, Real-Time Monitoring.

Introduction

Fire outbreaks continue to be one of the most devastating threats to residential homes, business

zones, and industrial facilities. The rapid spread of fires, the emission of dangerous gases, and structural damage result in significant financial loss and

endanger human lives. One of the primary causes of large-scale devastation in most instances is delayed detection, as well as slow response to emergencies. Traditional fire alarm systems are normally based on single-parameter detection, such as smoke detection, which might lead to false alarms or the inability to detect fire in its initial stages. Hence, a smart multi-parameter real-time monitoring system should be implemented to improve fire safety and minimize potential harm.

The booming development of the Internet of Things technology has made it possible to create smart monitoring systems that can observe and automatically control the environment. IoT systems are based on sensors, microcontrollers, and communication modules, which can offer real-time data acquisition and real-time alert generation. Using a combination of various sensing parameters, including smoke concentration, gas levels, and temperature change, it is possible to increase detection accuracy significantly. Such multi-sensing solution decreases the number of false alarms and provides dependable fire hazard detection.

This paper presents the proposal of an IoT-based Fire Detection and Alert System based on the ESP32 microcontroller with an MQ-2 smoke and gas sensor, a special gas detector, and a DHT22 temperature sensor. In the system, the environmental conditions are monitored continuously and sensor data analyzed to identify an abnormal level of smoke, dangerous gas concentration or a rapid increase of temperature. On the detection of fire related conditions, the system triggers multi-layered response systems allowing local alerts such as a buzzer and red light, automatic activation of water sprinklers to suppress fire, SMS messages to the nearby fire stations, and telephone messages to surrounding people to respond quickly.

The proposed system will utilize a modular and cost-effective architecture that is programmed in Arduino IDE hence reliable, easy to maintain and scalable. The combination of automatic suppression features and emergency communications, makes the system

appropriate to be used at home, office, laboratory and industrial settings. The system will greatly support fire safety and minimize the risk that can result to fire by allowing the detection of fire early and taking prompt measures.

RELATED WORK

With the rapid development of embedded systems and Internet of Things (IoT) technologies, fire detection systems have significantly evolved. Traditional fire alarm systems were primarily based on single-parameter sensing, such as standalone smoke detectors or temperature sensors, without real-time communication capabilities. Although these systems were simple and cost-effective, they suffered from delayed response and higher false alarm rates due to reliance on only one sensing parameter. These limitations encouraged researchers to explore multi-sensor and IoT-based intelligent fire detection solutions.

A proposed IoT-Based Smart Home and Office Fire Notification Alert System by Debnath et al. uses a set of temperature, smoke, and carbon monoxide sensors to identify the presence of fire as quickly as possible with the help of an Arduino microcontroller [1]. The system can also be applied as an environmental monitoring system that continuously transmits information in real time through the utilization of the MQTT protocol. The mobile application can be remotely accessed to monitor fire hazards from any location. The system also includes local alerts through buzzers and light indicators, along with an automatic sprinkler system to extinguish the fire at its initial stage.

Although the system improves the speed of the notification process and remote monitoring, it is highly dependent on internet connectivity and cloud communication [1]. Little is mentioned about multi-channel emergency communication, including direct alerts to fire stations. Besides, issues related to security architecture and scalability are not widely addressed,

which can affect deployment in large institutions or industrial applications.

Bhoi et al. proposed a multi-sensor FireDS-IoT system that uses a combination of MQ-135, MQ-2, MQ-7, and DHT-11 sensors along with an Arduino platform [2]. The system applies machine learning models, such as K-Nearest Neighbor (K-NN) and Decision Tree, to categorize environmental conditions as fire, no fire, and maybe fire. The research findings show a sufficiently high degree of classification accuracy, with K-NN achieving about 93 percent detection performance [2].

The data-driven approach significantly improves detection accuracy and minimizes false alarms; however, the system focuses more on analytical classification rather than real-time physical intervention [2]. It does not include automatic sprinkler-based suppression systems or comprehensive emergency communication mechanisms. Implementation in embedded systems may also become more complex due to reliance on dataset training and computational processing.

The second study aimed at enhancing previous smoke detector configurations by offering an integrated smoke detection and fire protection system using Cisco Packet Tracer.

Singh et al. proposed an Integrated Smoke Detection and Fire Prevention System that uses Cisco Packet Tracer to simulate interconnected safety devices, including smoke detectors, CO₂ sensors, sirens, and fire sprinklers [3]. The system focuses on network-based coordination of IoT-enabled devices and demonstrates how centralized control units can trigger alarms and sprinklers during hazardous conditions.

The study provides a good understanding of network-based fire safety architecture but is primarily simulation-based and does not demonstrate real-life hardware implementation [3]. Real environmental calibration, physical deployment validation, and GSM-based emergency alert integration are not elaborated upon. Therefore, the system remains

limited to conceptual and simulation environments rather than being field-ready.

The Home Automation System presented by Panda et al. is an IoT-Based system that focuses on LPG gas and fire detection with automated safety measures [4]. The system combines MQ2 gas sensors, DHT11 temperature sensors, servo-controlled gas valves, relay-operated exhaust fans, buzzer alerts, and SMS notifications through GSM. IoT platforms provide real-time monitoring and allow users to receive notifications and monitor environmental conditions remotely.

The system offers useful automated intervention, particularly in LPG leakage cases [4]. However, it is more focused on gas safety rather than comprehensive multi-parameter fire detection. Dedicated multi-sensor cross-validation for fire ignition detection is not a primary focus, and sprinkler-based fire suppression is not a major feature. Scalability and institutional deployment are also addressed only briefly.

Sivanesan et al. designed an IoT-Based Smart Forest Fire Detection and Prediction System that combines environmental sensors, an ESP32 microcontroller, GPS devices, and cloud-based machine learning models for real-time fire monitoring and predictive analytics [5]. The system not only detects fire conditions but also predicts possible fire spread patterns based on environmental parameters such as temperature, humidity, and wind speed.

Although the architecture is highly sophisticated and suitable for large-scale forest environments, it requires cloud access, continuous network connectivity, and significant computational resources [5]. The system is aimed at large ecological monitoring rather than small residential or industrial fire protection systems. Furthermore, localized automatic suppression mechanisms are not incorporated within the sensing architecture.

Gap in Research Findings

Based on the literature review [1]–[5], it can be noted that current systems mainly focus on cloud-based

monitoring, machine learning classification, LPG gas safety automation, network simulation, or large-scale predictive modeling. However, few studies combine multi-sensor fire detection, automatic water-based suppression, GSM-based emergency communication, and localized alerts within a compact, modular, and cost-effective hardware platform.

The proposed IoT-Based Fire Detection and Alert System addresses this gap by integrating smoke sensors, gas sensors, temperature sensors, automatic sprinkler activation, buzzer and LED alarms, SMS alerts to fire stations, and phone alerts to nearby individuals within a unified multi-layered response system.

METHODOLOGY

The IoT-Based Fire Detection and Alert System is meant to be a well-organized multi-parameter monitoring and automated responding system. The methodology also provides a continuous environmental monitoring, correct identification of hazards, and prevention activation. It combines sensory elements, on-board processing and communication units to create a full fire safety solution. The overall strategy is aimed at minimizing the delay in detection, the occurrence of the false

alarm, and offering a multi-layered emergency response. The system has a clear sensing, processing, validation and response sequence in order to maintain a stable operation in the various environmental conditions. It enhances the accuracy of detection and increases safety by measuring smoke, gas and temperature. Such a combined approach makes sure that alerts are timely, suppression is effective, and communication is reliable, which makes the overall fire safety control more efficient.

A. Overall System Design:

The system architecture follows a modular layered framework consisting of three primary layers: sensing layer, processing layer, and response layer. The sensing layer continuously monitors environmental parameters such as smoke density, gas concentration, and ambient temperature. The processing layer analyzes the collected sensor data and determines hazard conditions. The response layer activates alert mechanisms, suppression systems, and emergency communication protocols.

This modular design ensures flexibility, scalability, and ease of maintenance. Each module operates independently while contributing to the overall system functionality, thereby enhancing reliability and performance.

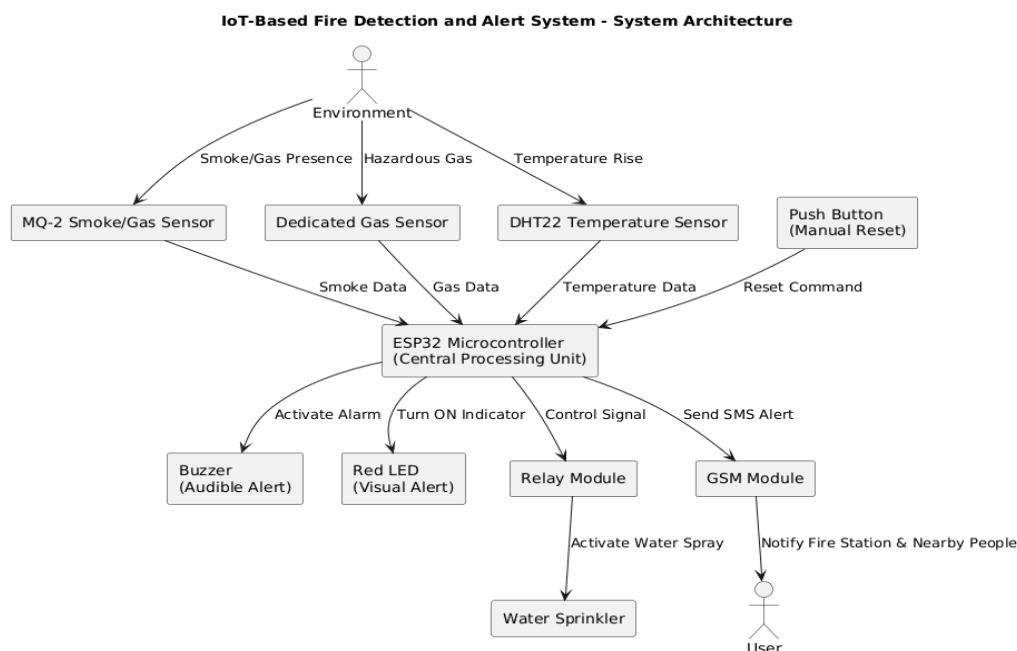


Fig:1 System Architecture

B. Multi-Sensor Environmental Monitoring:

The sensing mechanism incorporates the following sensors; an MQ-2 smoke and gas sensor, a dedicated gas sensor, and a DHT22 temperature sensor. The MQ-2 sensor is used to detect combustible gases and smoke particles and gives an analogous response based on the levels of concentration of the gases. The extra special gas detector is sensitive to certain dangerous gases and, hence, increases the accuracy of detection. The DHT22 sensor monitors the ambient temperature and detects the abnormal rise of temperature that is related to fire ignition.

Multiple sensing parameters are used to provide redundancy and cross-checking which greatly minimises false triggering. This is done by regularly performing continuous monitoring in order to ensure real time responsiveness and the sensor information is then sent to the ESP32 microcontroller to be analyzed. The system receives a combination of smoke and gas measurements along with temperature which enhances the reliability of detection and enhances detection of the early fire hazard.

C. Programmable and Dynamic Processing and Decision-Making:

The ESP32 microcontroller functions as the central processing unit of the system. It continuously reads sensor outputs and converts analog signals into digital values for analysis. Predefined threshold values are established based on safe environmental limits for smoke, gas concentration, and temperature.

The hazard detection logic operates through the following sequence:

1. Sensor value acquisition.
2. Comparison with predefined threshold values.
3. Multi-parameter validation to confirm fire conditions.
4. Decision flag generation for emergency response activation.

This threshold-based decision mechanism is computationally efficient and ensures minimal latency. The ESP32 platform provides sufficient processing capability and integrated communication

support, making it suitable for real-time embedded applications.

D. Multi-Layered Alert Mechanism:

Definition: Multi-Layered Alert Mechanism is the use of numerous warning and response mechanisms to activate in order to detect, alert and suppress fire hazards quickly.

When the system detects the existence of hazardous conditions, it switches on audible and visual alerts promptly. To alert occupants a high intensity sound signal is produced by a buzzer and a visible warning signal is produced by a red LED. These local alerts assure real time awareness in the environment under surveillance.

The activation of the alert process is performed immediately after the confirmation of the hazard, and therefore the response is fast in the case of critical situations.

E. Automatic Fire Suppression:

In addition to the alert generation the system will also be equipped with automatic water sprinkler system which would be controlled through a relay module. A fire hazard is sensed when the sprinkler is switched on by means of the relay to enable water to be released to put out the fire in its early days. This automatic measure assists in regulating and containing the spread of fire before it turns out to be a huge event. The system reduces the reliance on human response as suppression is initiated when the hazard has been confirmed thus minimizing damage.

The system includes suppression and detection and this is a major improvement of the system as it makes performance significantly better than traditional alert-only fire alarm systems. The proposed system also takes action unlike a traditional system which merely notifies the occupants since they have sensed the hazard. The sprinkler system is a relay-controlled device that provides quick physical intervention, which consequently reduces the intensity of the fire, structural, and environmental damage. This type of detection and suppression combination is a way of

improving safety performance and the fire risk reduction in general.

F. Emergency Communication Protocol:

The The system has a GSM communication module that will help provide a reliable and real-time transmission of alerts to ensure an external response to an emergency. When a fire hazard has been confirmed, the ESP32 microcontroller will trigger a series of commands to the GSM module to send SMS alerts to nearby fire stations and provide phone alerts to people surrounding the area. This mechanism of communication through many channels makes sure that the information about the emergencies is relayed to the responsible authorities and the affected individuals in time.

The GSM module works based on the cellular network infrastructure that can ensure the reliability of communication as opposed to the systems that depend on the internet. The system will not depend on cloud platforms or Wi-Fi connectivity and therefore will not disrupt the delivery of alerts in an environment with limited or unstable internet connectivity. Most vital alert information is included in the emergency message, which allows swift situation awareness and coordinated action.

The communication protocol is created to act as soon as the hazard validation is performed in order to reduce the response time. Automated transmission process excludes the use of manual intervention and alerts are received even when occupants do not know about the incident at all. The local detection and external communication is an integration that will enhance the emergency preparedness and response efficiency.

Moreover, institutional response is reinforced by direct messaging to fire stations, and the community-level awareness and quick help is provided through alerts to nearby people. This formal communication plan is more effective in improving the overall performance of the fire detecting system because it incorporates both local response in the short-term and external response in the short-term.

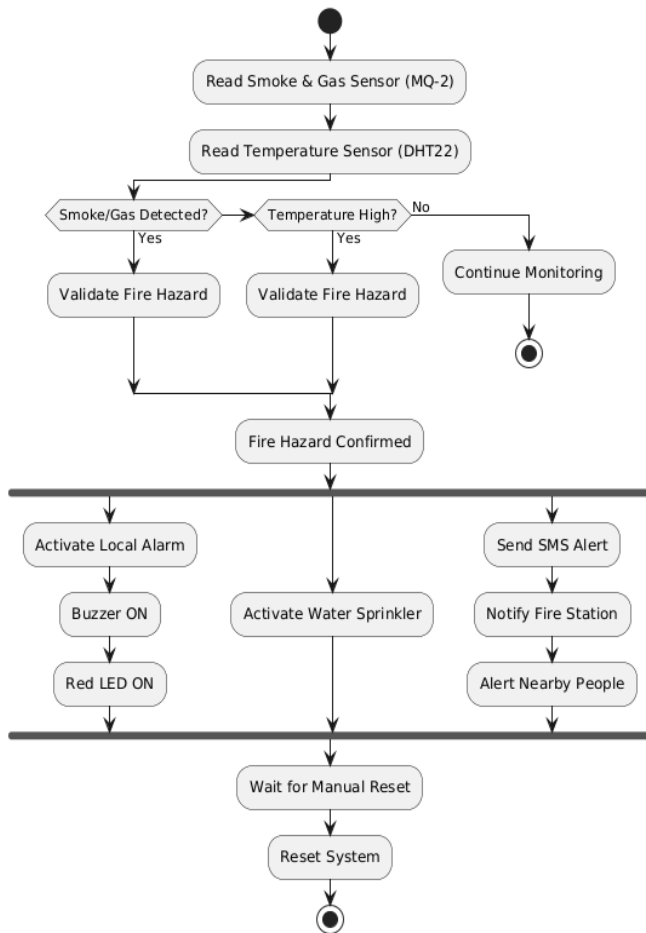
G. Manual Reset and False Alarm Control:

There is also a manual push button so that the users can disable or re-set the system in case of false alarms. This aspect enhances convenience and avoids unjustified panic but does not compromise the safety of operations. The system automatically goes back to normal monitor mode after restarting and therefore provides protection

H. Software Implementation and System Flow:

The entire system will be programmed through the Arduino IDE. The entrenched firmware includes sensor initiation programs, pervasive watch logic, comparison of thresholds, alert trigger and sprinkler control programs and the GSM communication programs. The program code is designed in such a way that it consumes minimum memory resources although it is stable and reliable in its execution.

The general system working is formed of the constant control, recognition of hazard, activation of the alarms, sprinkler control, notification of the emergency and manual re-setting. This is a methodical manner of responding in time and detecting fire hazard promptly and reducing fire hazards. Sensitivity tests are also fitted to ensure correct measurements in diverse environmental conditions. The error-handling systems are also put in place to promote the continuous running of the system when things go out of shape.

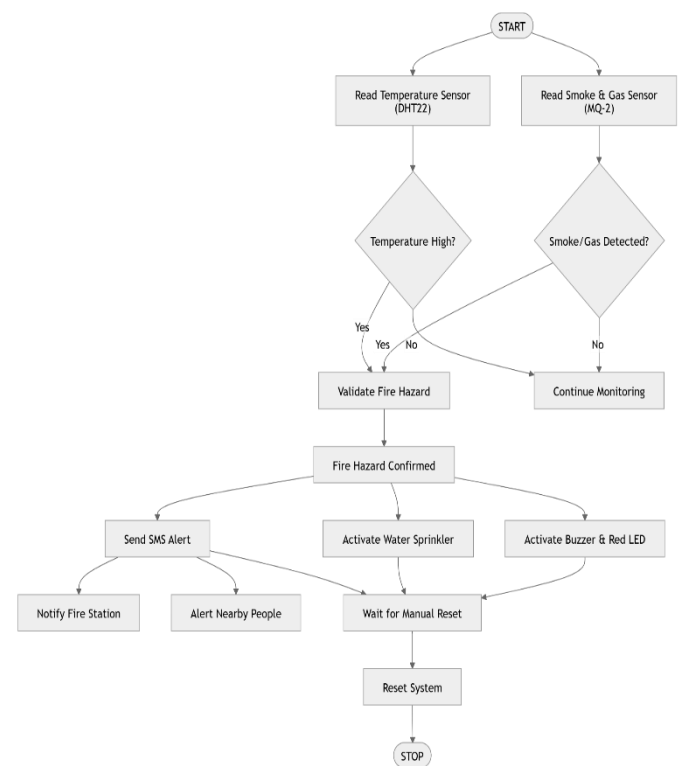
IoT-Based Fire Detection and Alert System - Flowchart**Fig 2: Flowchart**

The methodology used in working of the proposed IoT-Based Fire Detection and Alert System is aimed at providing uninterrupted control, quick hazard detection, and activation of response. The system works in a systematic order starting with real time environmental monitoring with the MQ-2 smoke sensor, specialized gas sensor and the DHT22 temperature sensor. These sensors constantly gather the data of the smoke concentration, gases level and ambient temperature and send the data to ESP32 microcontroller where it is processed.

The sensor values are compared to set safety limits to deduce the presence of abnormal conditions by the ESP32. This is done to ensure multi-parameter validation is an indicator of the presence of a fire hazard, thus minimizing false alarms. After a dangerous situation has been determined, the system

will automatically trigger local warning systems such as a buzzer and red LED. At the same time, the relay module causes the automatic sprinkler of the house water system to turn on and extinguish the fire at its initial stage.

Along with the local response, the GSM module is turned to provide the SMS alert to the nearby fire stations and emergency contacts to make sure that external help is provided. There is a manual push button to enable the user to restart the system in case of false alarms. Once response cycle is over the system automatically goes back to monitoring mode and does not allow any discontinuation of fire protection.

**Fig 3: Working Methodology**

RESULTS/FINDINGS

The proposed IoT-Based Fire Detection and Alert System was tested under different controlled environmental conditions to evaluate its performance, reliability, and response efficiency. Testing was carried out by simulating the presence of smoke, gas leakage, and temperature increase to assess the system in terms of detection sensitivity and response time.

During the simulation of smoke using controlled combustible material, the MQ-2 sensor was capable of detecting abnormal smoke concentration and transmitting the data to the ESP32 microcontroller. Similarly, a higher temperature environment was simulated to test the capability of the DHT22 sensor to respond to rapid temperature rise. The additional gas sensor was effective in detecting the presence of hazardous gases, thereby confirming the multi-sensor integration methodology. The system demonstrated accurate detection with minimal false triggering due to multi-parameter validation.

The system provided clear visual and audible alerts through activation of the buzzer and red LED when a hazard was confirmed. The relay module successfully activated the automatic water sprinkler system, enabling early fire suppression. Additionally, the GSM module sent SMS messages to predefined emergency contacts and nearby fire stations. The average response time of the system from hazard detection to alert activation was observed to be a few seconds, ensuring rapid response.

The findings validate that the combined use of smoke, gas, and temperature detection increases detection accuracy and minimizes false alarms compared to single-sensor systems. The multi-layered response mechanism integrating alert generation, automatic suppression, and emergency communication significantly enhances fire safety performance. Overall, the system was found to be stable, reliable in operation, and effective in hazard mitigation under various testing conditions.

DISCUSSION

The The Fire Detection and Alert System, an IoT-Based Fire Detection and Alert System, was applied and experimented in order to prove the efficiency of the integration of several sensing parameters in the early detection of fires. The solution is better than the traditional single sensor fire alarm systems as it uses smoke, gas, and temperature to assist in detection and minimizing false alarms. Fire hazards are verified

with the help of the multi-parameter validation method, and cross-verification leads to the increased reliability of the system in various environmental conditions.

The automatic fire suppression through the provision of a water sprinkler system that is controlled by a relay is very important in improving the safety performance of the system. Whereas the traditional fire alarm systems are largely aimed at the generation of alerts, the proposed system offers instant physical intervention besides the alert messages. Such a conglomerate strategy of identification and extinction assists in lowering the dissemination of the fire at the initial phase, which minimizes possible structural harm and hazards.

The practicality of the system is also enhanced by the fact that the GSM-based emergency communication has been included. This system is able to help in a quick external response by sending SMS alerts to the local fire stations and predetermined contacts. The system of communication is not based on the internet-based platforms and is thus applicable in the circumstances where internet connectivity cannot be trusted. This increases its reliability in contrast to cloud-based systems.

In sum, the proposed system possesses a steady working performance, minimal response time, and effective hazard mitigation. It is modular in design and affordable, simple to install and maintain and could be applicable in residential, commercial and industrial settings. The outcomes of the experiment demonstrate that multi-sensor surveillance combined with automated suppression and emergency communication make up a system of fire safety, which is effective and holistic. The combination of detection, alert and suppression strategies ensures that the evocation of fire emergencies is fast and synchronized. In general, the system can be characterized by a high level of reliability.

CONCLUSION AND FUTURE SCOPE

The suggested IoT-Based Fire Detection and Alert System can be a helpful and trustworthy platform to detect fire early and be able to react quickly in an emergency. The system will have smoke, gas, and temperature sensors, which are combined with an ESP32 microcontroller and promise to provide constant environmental monitoring and correct hazard identification. Multi-parameter validation enhances reliability in detecting as well as reducing false alarms. Along with audible and visual notifications, the system will activate the automatic water sprinkler system and send the GSM-based alerts to fire stations and emergency contacts which can form a multi-layered approach towards responding to the fire. The effectiveness of fire hazard mitigation within different conditions is proved by experimental testing that ensures the stable operation and rapid reaction time.

To achieve a better solution, the remote supervision and data logging might be achieved in real-time with the help of cloud-based monitoring. The use of sophisticated analytics or intelligent prediction models would be beneficial in the evaluation of risks in advance. More features such as support to mobile applications, camera based tracking and sophisticated sensor tuning can also enhance usability and accuracy. Lastly, it would be beneficial to scale the system to large-scale industrial use with centralized monitoring functions to enhance its application in smart fire safety.

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